

Rohith Reddy Kolanu

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[Portfolio](#)

Summary

Data Science and Analytics professional with 2 years of experience in data analysis, automation, and business intelligence at Broadridge. Proficient in Python, SQL, Power BI, Statistical Analysis, Machine Learning, Neural Networks, NLP, and Generative AI. Built predictive models, automated reporting, and bridged business strategy with data-driven solutions.

Education

The University of Texas at Austin

Austin, TX

Post Graduate Program in Artificial Intelligence and Machine Learning, CGPA: 4.33

Dec 2025 - Aug 2026

Relevant Coursework: Python Foundations, Machine Learning, Advanced Machine Learning, Introduction to Neural Networks, Natural Language Processing with Generative AI, Introduction to Computer Vision, Model Deployment

University of Massachusetts Amherst

Amherst, MA

Master of Science in Business Analytics, Data Analytics, CGPA: 4.0

Aug 2024 - May 2026

Relevant Coursework: Data Management, Business Intelligence, Business Applications with Python, Machine Learning, Project Management, Financial Technology, Data Science, Data Visualization, Artificial Intelligence

Experience

Broadridge

Hyderabad, India

Process Analyst

Aug 2022 - Jul 2024

- Designed and automated an Excel-based utilization tracking system, reducing manual effort by 90 percent, standardizing calculations, and enabling accurate workforce analytics leveraged through monthly Power BI and PowerPoint reports.
- Built and delivered monthly performance dashboards using Power BI and Excel, improving reporting accuracy and visibility of key utilization metrics, trends, and KPIs to support data-driven operational decisions.
- Analyzed and processed over 2,000 client documents, websites, and HTML emails with 99% accuracy, ensuring SLA compliance while maintaining high data quality, validation standards, and process consistency.
- Handled 300+ operational requests during peak cycles with 99% accuracy, reducing turnaround time by 20% through workflow optimization, prioritization logic, and continuous process improvement initiatives.
- Collaborated cross-functionally with Desktop Publishing Group and Client-Facing team to validate data outputs, improve reporting reliability, and translate operational data into actionable business insights.

Projects

Optimizing Wind Turbine Maintenance Costs Through Neural Network-Based Failure Prediction | Python

- Built an end-to-end wind turbine failure prediction pipeline on 25,000 records with 40 ciphered sensor features, trained and tested 9 neural networks with regularized Model 7 achieving 98.52% accuracy and 0.931 F1-score on test data.
- Engineered a preprocessing pipeline with median imputation, standardization, SMOTE class balancing on 17:1 imbalance, and cost-based threshold tuning, optimizing recall to capture 93.21% of failures and minimize replacement costs.

Airbnb Price Prediction, Guest Sentiment Analysis & Topic Modeling using ML & NLP | TensorFlow, Keras

- Built an end-to-end Airbnb price prediction pipeline on 3,840 Chicago listings using structural and NLP features, trained and tested 2 neural networks and tuned XGBoost achieving 0.617 test R-squared and \$81.23 RMSE as the top model.
- Engineered an NLP pipeline on 271,871 reviews with VADER, DistilBERT sentiment, zero-shot scoring across 5 categories, and LDA, NMF, LSA, BERTopic modeling, turning unstructured feedback into listing-level pricing signals.

Predicting Visa Approval Outcomes Using Advanced Machine Learning Techniques | Python, Scikit-learn

- Built an end-to-end visa approval prediction pipeline on 25,480 records using applicant, employer, and wage features, tested 5 ensemble classifiers with tuned XGBoost achieving 74.25% accuracy and 0.819 F1-score as the top performer.
- Engineered a preprocessing pipeline with one-hot encoding, SMOTE-based class balancing, stratified splitting, and hyperparameter tuning, optimizing recall to 87.31% to capture certified applications and streamline OFLC screening.

Technical Skills

Programming Languages & Databases: Python, R, SQL, MSSQL, PostgreSQL, MySQL, NoSQL, Git

Data Analysis & Visualization: Pandas, NumPy, SciPy, Matplotlib, Seaborn, Plotly Express, Excel, Power BI, Tableau

Machine Learning & AI: Scikit-learn, Regression, Classification, Clustering, Deep Learning, Neural Networks, NLP

Deep Learning & GenAI: TensorFlow, Keras, PyTorch, LLMs, Hugging Face, LangChain, ChromaDB

Big Data & Cloud Platforms: Apache Spark, Kafka, Airflow, Hadoop, ETL/ELT Pipelines, Snowflake, AWS

Model Deployment & MLOps: Docker, Flask, Google Colab, Jupyter Notebook, OpenCV, spaCy